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Ion microprobe baddeleyite and zircon ages for Late Archaean mafic dykes of the Pilbara Craton, Western Australia*

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Ion microprobe U–Th–Pb analyses of baddeleyite and zircon yield precise ages for several mafic intrusions in the Pilbara Craton of Western Australia. Baddeleyite was dated from four dolerite dykes of the north-northeast-trending Black Range swarm intruded into granitoid–greenstone basement in the northern part of the craton. The mean $^{207}Pb^*/^{206}Pb^*$ age of 2772 ± 2 Ma, interpreted as an unambiguous age of emplacement for the dykes, is within error of previous ion microprobe U–Pb zircon ages for the Mt Roe flood basalts and confirms that the dykes acted as feeders to the volcanic rocks. The Sylvania Inlier, in the southeastern Pilbara Craton, also contains north-northeast-trending dykes that were correlated previously with the Black Range swarm. Based on concordant and discordant zircon analyses from samples of two dykes, the best estimate of the age of the Sylvania dykes is 2747 ± 4 Ma. The Sylvania dykes thus appear to be significantly younger than, and hence unrelated to, the Black Range swarm, but may have acted as feeders to younger volcanic units in the Fortescue Group such as the Kylena Formation.

Key words: Archaean, baddeleyite, dolerite, dyke, Pilbara Craton, radiometric dating, zircon.

INTRODUCTION

Granite–greenstone basement in the northeastern Pilbara Craton in northern Western Australia (Figure 1) is intruded by an extensive suite of mafic dykes known as the Black Range dyke swarm. The age of the Black Range swarm and its relationship to mafic volcanic successions in the region have been topics of discussion among Pilbara geologists for many years, as described below. The Sylvania Inlier, a granite–greenstone terrane in the southeastern Pilbara Craton, also contains north-northeast-trending mafic dykes (Suite 1 dykes of Tyler 1991) that were correlated with the Black Range swarm by Tyler (1991). This paper presents ion microprobe U–Th–Pb analyses of baddeleyite and zircon from dykes in the two areas. The results are interpreted as unambiguous ages of emplacement for these intrusions and help to resolve controversy regarding their correlation and affinities.

GEOLOGICAL BACKGROUND

In the northeastern part of the Pilbara Craton, *ca* 3.6 to *ca* 2.8 Ga granite–greenstone terranes (Blake & McNaughton 1984; Bickle *et al*. 1989; Williams & Collins 1990; Nelson 1998) are intruded by subparallel north-northeast-trending dykes of the Black Range swarm (Figure 2). They range from metre-scale dykes of aphanitic basalt to 100 m-wide dykes of coarse-grained granophyric quartz dolerite or gabbro, and are metamorphosed to sub-greenschist facies (Hickman 1978; Smith *et al*. 1982). Described first by Gregory (1861), the dyke swarm was named for the conspicuous 'dark volcanic ridges', ~60 m high, formed by a single intrusion known as the Black Range Dyke, which is more than 100 km long, up to 150 m wide and intrudes the Shaw Granitoid Complex southwest of Marble Bar (Noldart & Wyatt 1962; Bickle *et al*. 1983, 1989; Hickman 1983). A second prominent member of the swarm is the Cajuput Dyke, east of Nullagine, which is at least 60 km long and up to 1 km wide (Hickman 1978). Granite–greenstone basement rocks are overlain unconformably by flat-lying to gently folded volcanic and sedimentary strata of the Fortescue Group (Figure 2). These cover rocks are exposed as outliers on the northern Pilbara Craton and as continuous outcrop along its southern margin (Trendall 1990).

Black Range dykes are not generally in contact with, nor observed with certainty to intrude, the Mt Roe Basalt at the base of the Fortescue Group (Hickman 1983). Although Maitland (1906) considered the dykes to be younger than the 'Nullagine Beds' of the lower Fortescue Group, Noldart and Wyatt (1962) assigned a pre-Fortescue, Palaeoproterozoic age to the Black Range swarm. Lewis *et al*. (1975) reported a Rb–Sr isochron age for the Black Range Dyke of 2287±89 Ma (2σ), now regarded as anomalously young in light of recent U–Pb geochronology (described below). Lewis *et al*. (1975) also described the composition of the dyke and its metamorphic effects on the enclosing granite and proposed

*Tables 2–4 [indicated by an asterisk (*) in the text and listed at the end of the paper] are Supplementary Papers lodged with the National Library of Australia (Manuscript Section); copies may be obtained from the Business Manager, Geological Society of Australia.

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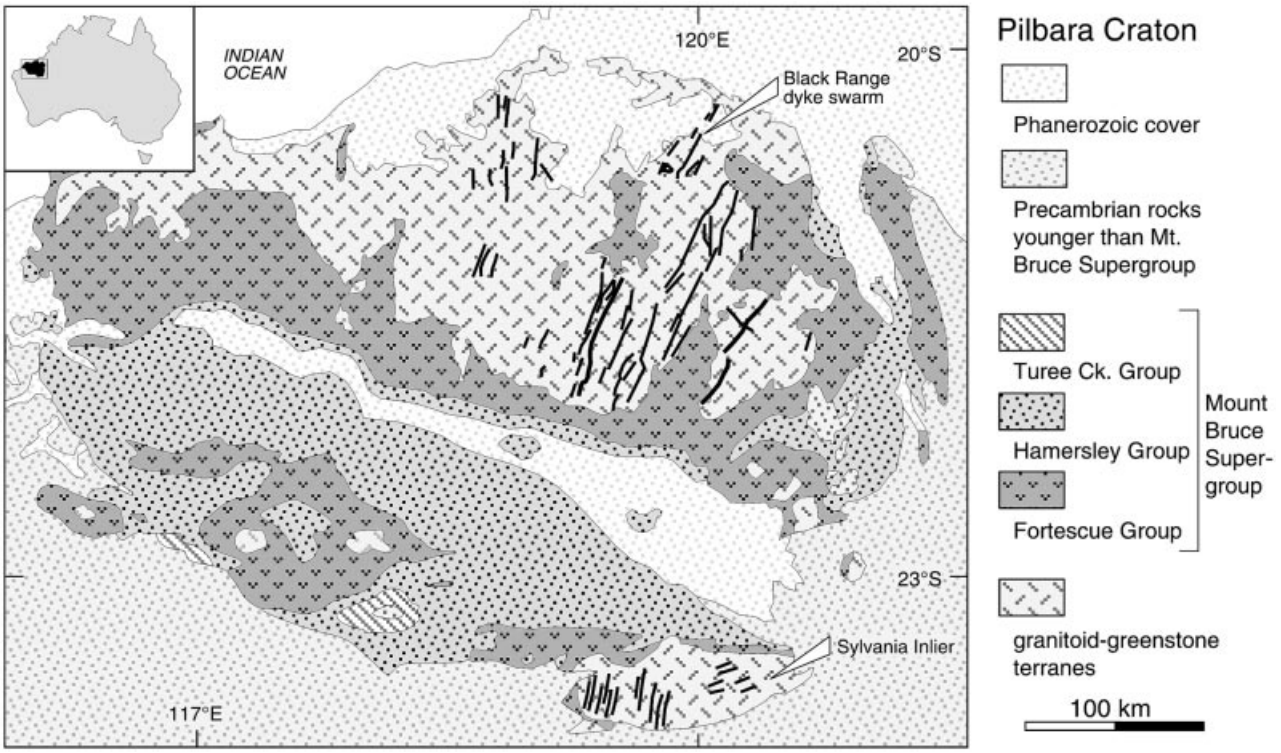


Figure 1 Simplified geology of the Pilbara Craton (after Wingate 1998). Dykes of the Black Range swarm and in the Sylvania Inlier are represented by black lines.

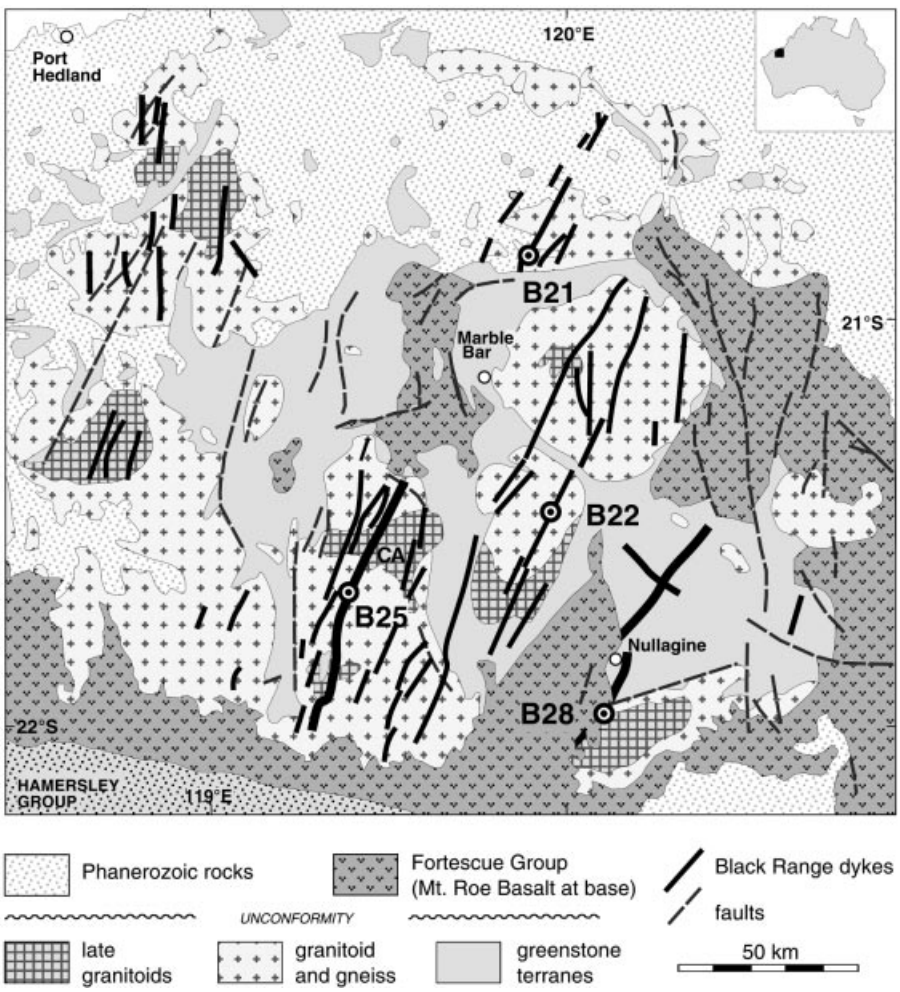


Figure 2 Geological map of the northeastern part of the Pilbara Craton showing the distribution of granite–greenstone terranes, Black Range dykes (and sample sites therein), and cratonic cover rocks of the Fortescue Group (modified from Hickman 1983). Thicknesses of dykes and faults have been exaggerated for clarity. CA, Cooglegong Adamellite.

that it was a feeder to the Mt Roe flood basalts. This view was advocated also by Hickman (1983), based on extensive field observations.

An older limit of 2851 ± 2 Ma (95% confidence limits) for emplacement of the Black Range Dyke is furnished by an ion microprobe U–Pb zircon age for the Cooglegong Adamellite (CA in Figures 2, 3), which the dyke intrudes (Nelson 1998). A younger limit for the swarm is provided by precise zircon U–Pb ages between 2764 and 2757 Ma for the Gregory Granitic Complex and coeval felsic igneous rocks in the Hardey Formation, which unconformably overlies the Cajuput Dyke (Arndt *et al.* 1991; Nelson 1996; Williams & Trendall 1998a, b). Palaeomagnetic poles obtained from the Black Range Dyke and the Cajuput Dyke are not significantly different, allowing these two intrusions to be approximately the same age (Embleton 1978). As noted above, no Black Range dykes are observed unequivocally to have intruded the Mt Roe Basalt (Hickman 1983); ion microprobe U–Pb zircon ages (2σ) of 2775 ± 10 and 2763 ± 13 Ma were obtained for felsic volcanic rocks near the base of the unit by Arndt *et al.* (1991) (Table 1).

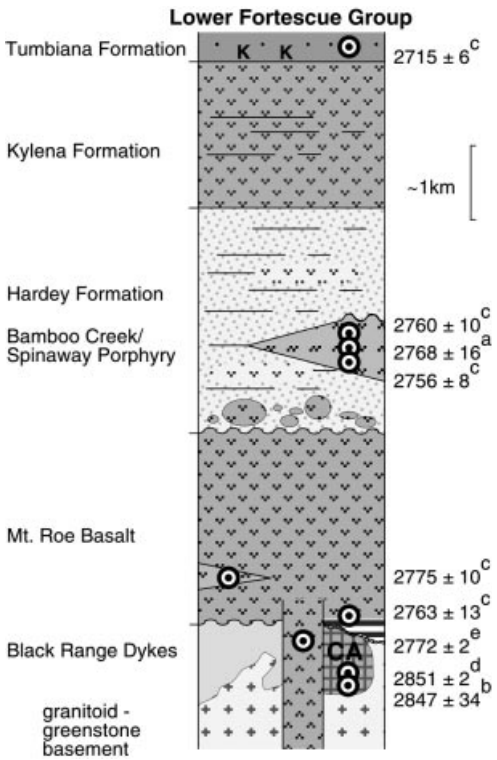


Figure 3 Idealised stratigraphic column showing radiometric ages (±2σ) and relationships between rock units in the lower Fortescue Group. References: a, Pidgeon (1984); b, Bickle *et al.* (1989); c, Arndt *et al.* (1991); d, Nelson (1998); e, this study. CA, Cooglegong Adamellite; K, komatiitic basalt.

The Sylvania Inlier in the southeastern Pilbara Craton (Figure 4) also contains north-northeast-trending mafic dykes that typically are medium to coarse grained and range up to 200 m or more in width (Suite 1 dykes of Tyler 1991). Dykes in the eastern part of the inlier have been rotated by younger deformation into an east-northeast trend (Tyler 1991). Mineral assemblages and textures are consistent with mainly static metamorphism to upper greenschist or lower amphibolite grade, thought to have resulted from burial beneath rocks of the Hamersley Basin (Smith *et al.* 1982; Tyler 1991). These dykes, referred to in this paper as the Sylvania dyke swarm, were correlated with the Black Range dykes by Tyler (1990, 1991) and Tyler *et al.* (1992). The latter authors suggested, based on differences in REE, Rb–Sr and Sm–Nd systematics, that granitoid rocks of the Sylvania Inlier and the northern Pilbara craton were derived from different sources. They proposed that the two regions represent separate tectonostratigraphic terranes that were amalgamated between 3.0 and 2.76 Ga, and subsequently 'stitched together' by dykes of the Black Range swarm. No previous geochronology has been undertaken on the Sylvania dykes.

SAMPLE DESCRIPTIONS

Samples were obtained from fresh surface exposures of the coarsest grained, most central portions of the dykes (GPS coordinates for all samples are provided in Tables 2*–4*).

Table 1 Mean $^{207}Pb^*/^{206}Pb^*$ ages for the Black Range and Sylvania dykes and Mt Roe Basalt.

Sample	*n*	mean age	standard error (1σ)	95% error
Black Range dykes				
B21	21	2768.0	2.7	5.6
B22	24	2774.7	2.2	4.6
B25	22	2771.6	2.3	4.8
B28	15	2771.5	2.0	4.3
mean	–	2771.8	1.1	2.3
Mt Roe Basalt				
86738	11	2775.3	4.4	9.8
94792	11	2762.5	5.8	12.5
mean	–	2770.6	3.5	7.3
Sylvania dykes				
Z30	8	2749.6	2.5	5.9
Z31	19	2745.1	2.2	4.6
mean	–	2747.1	1.7	3.5

n, number of analyses averaged. Mt Roe Basalt data from Arndt *et al.* (1991).

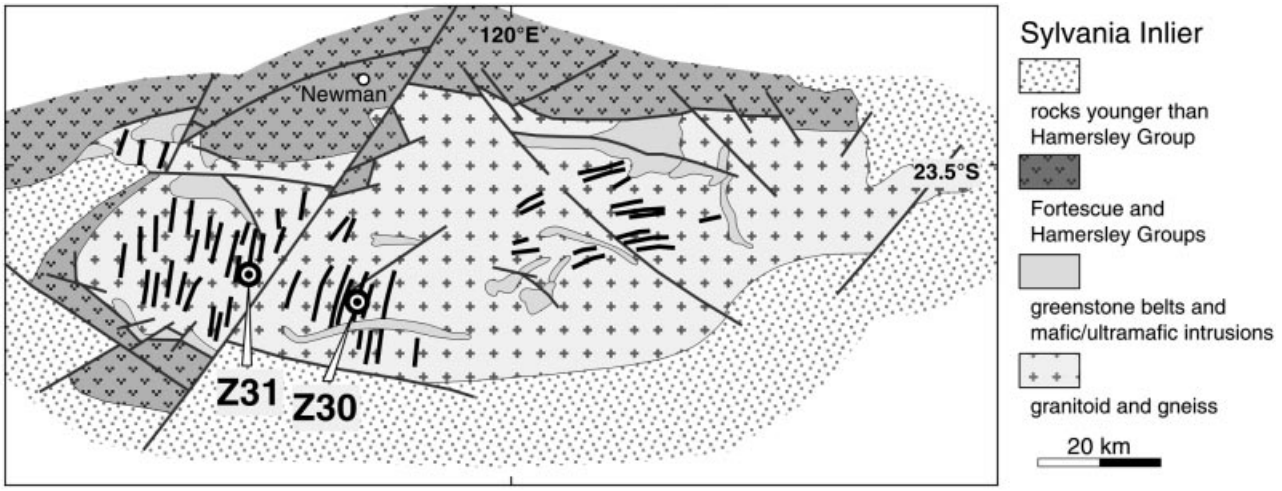


Figure 4 Geology of the Sylvania Inlier (simplified from Tyler 1991) showing locations of samples Z30 and Z31. Black lines represent Archaean dykes of the Sylvania swarm (suite 1 of Tyler 1991); grey lines represent faults. Thicknesses of dykes and faults are exaggerated.

The four Black Range samples are similar petrographically and exhibit subophitic to gabbroic textures. Microscopic examination reveals prismatic labradorite (40–50%) and subhedral to euhedral augite (30–40%), with granophyric intergrowths of quartz and feldspar (15–20%) radiating from plagioclase laths and filling interstices. Alteration is moderate to advanced, particularly along cleavage traces and fractures, with plagioclase partially or completely saussuritised, and augite crystals partially replaced by chlorite and actinolite. Also present are minor amounts of apatite, epidote, K-feldspar, ilmenite and magnetite. Pleochroic brown baddeleyite was noted within chlorite and augite.

To recover as many large and intact crystals as possible for analysis using the SHRIMP ion microprobe, baddeleyite was extracted by dissolving the silicate minerals of each sample in hydrofluoric acid (HF) over a period of three days. It was demonstrated by Wingate (1997) that immersion of baddeleyite in HF for up to seven days has no detectable effect on Pb isotopic ratios measured in baddeleyite by ion microprobe. Between 30 and 60 euhedral baddeleyite crystals and crystal fragments were separated from ~350 g of each Black Range sample. Most crystals are less than 100 μm long, although several range up to 100×300 μm. No zircons were recovered from the Black Range dyke samples.

Samples Z30 and Z31 were collected from medium-grained, north-trending dykes in the western part of the Sylvania Inlier. Both samples have been metamorphosed to lower amphibolite facies and are variably recrystallised, although igneous features are generally preserved. Z30 was obtained from the centre of a prominently exposed, 15 m-wide dyke. The sample is composed mainly of poikilitic green and blue-green amphibole (40%), quartz (35%), and sericitised plagioclase (15%). Sphene (5%) is found both as euhedral crystals and rimming iron oxides. Euhedral and subhedral zircon is prominent within pleochroic haloes in amphibole. Sample Z31 was collected from loose, rubbly outcrop in a 5 m-wide dyke. It is similar in character to Z30 but contains more amphibole (45%) and plagioclase (45%), and much less quartz (5–10%) and zircon. Plagioclase is less sericitised than in Z30, and sphene (5%) is again present as euhedral crystals and rimming iron oxides. Zircons in Z30 are variably discoloured and cracked; in both samples, zircons contain abundant, unidentified sub-micrometre size inclusions. Zircon from the two Sylvania samples was extracted using conventional techniques, without HF. Although zircon will survive the HF separation technique, the effects of HF on the U–Th–Pb systematics of zircon have not been investigated.

SHRIMP GEOCHRONOLOGY

U–Th–Pb measurements were conducted using the SHRIMP II ion microprobe at the Research School of Earth Sciences, Australian National University, and employed standard operating and data-processing procedures (Compston *et al.* 1984, 1992; Claoué-Long *et al.* 1995; Wingate *et al.* 1998). Owing to crystal orientation effects that bias $^{206}Pb/^{238}U$ ratios measured in baddeleyite by ion microprobe (Wingate 1997; M. T. D. Wingate & W. Compston unpubl. data), only $^{207}Pb^*/^{206}Pb^*$ data are reported here (Pb* = radiogenic Pb). Accuracy of measured $^{207}Pb^*/^{206}Pb^*$ ratios was monitored by repeated analysis of a baddeleyite reference standard characterised previously by isotope dilution thermal ionisation mass spectrometry (IDTIMS); in all cases measured ratios were indistinguishable statistically from IDTIMS results. Values for U and Th concentration in baddeleyite are approximate owing in part to the lack of a suitably homogeneous reference baddeleyite, although they are correct relative to each other within a single analytical session. Baddeleyite analyses were corrected for common Pb using the 208-method (Hinthorne *et al.* 1979; Wingate *et al.* 1998). Common Pb correction in zircons employed non-radiogenic ^{204}Pb. In both cases, an average crustal common Pb composition appropriate to the age of the mineral was assumed, although almost all corrections are sufficiently small to be insensitive to the choice of common Pb composition. Absolute U and Th concentrations and $^{206}Pb^*/^{238}U$ ratios in zircons from the Sylvania dykes were determined relative to the SL13 zircon standard (Claoué-Long *et al.* 1995).

Black Range dykes

Baddeleyite from Black Range samples was analysed during two analytical sessions (Tables 2*, 3*). Apparent uranium concentrations range from 170 to 5600 ppm, and $^{232}Th/^{238}U$ ratios range from < 0.01 to 0.24. Values for $f206$ (the proportion of common to total ^{206}Pb) are much less than 1%, so that in general errors due to common Pb correction are insignificant. Within each sample, all $^{207}Pb^*/^{206}Pb^*$ ratios agree to within analytical precision and are approximately normally distributed, and the weighted means for each sample also agree to within error. Low dispersion within results for each sample indicates that the crystals experienced no significant loss of radiogenic Pb, or that Pb loss if present occurred recently. The results are summarised in Table 1 and Figure 5. The overall mean age for four samples is 2772 ± 2 Ma (95% confidence limits).

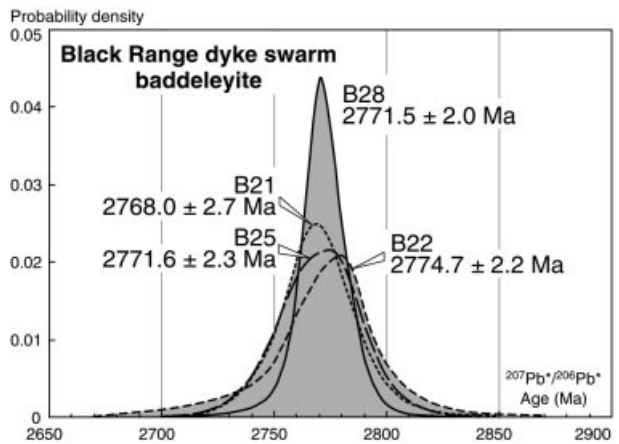


Figure 5 Probability density plot showing the distribution of $^{207}Pb^*/^{206}Pb^*$ ages for baddeleyite from four Black Range dykes. Errors on mean ages are ±1σ.

Sylvania dykes

Zircons from samples Z30 and Z31 were analysed during separate sessions (Table 4*). For sample Z30, 23 analyses were conducted of 19 zircons, and for Z31, 23 analyses were conducted of 22 zircons. Uranium concentrations are similar for both samples, ranging from 125 to 570 ppm. Th/U ratios in Z30 zircons range from 0.6 to 1.3 (two at 0.2), and in Z31 range from 1.2 to 2.8. In Z30, values for $f206$ are less than 0.9%, averaging 0.4%; in Z31, $f206$ is less than 0.5% for all but one analysis, and averages less than 0.2%.

The pattern of dispersion for Z30 indicates that the zircons have experienced both ancient and recent loss of radiogenic Pb (Figure 6). Although this scatter precludes regression of the data to yield Concordia intercepts, the group of eight analyses with the highest $^{207}Pb^*/^{206}Pb^*$ ratios yields a mean $^{207}Pb^*/^{206}Pb^*$ age of 2749.6 ± 2.5 Ma (1σ). It cannot be assumed from these data that zircons in this group experienced only geologically recent Pb loss, hence this must be considered a minimum age for crystallisation of the zircons. As with Z30, analyses of Z31 zircons indicate both ancient and recent Pb loss, but to a much lesser degree (Figure 6). Although four analyses are significantly reversely discordant and two are strongly normally discordant, the majority are within error of Concordia. Assuming only recent Pb loss, the weighted mean $^{207}Pb^*/^{206}Pb^*$ age is 2742.1 ± 1.9 Ma, whereas regression of all data yields Concordia intercepts of 2740.8 ± 2.4 and 313 ± 102 Ma (all ± 1σ).

Within a single igneous population, zircons that yield the highest $^{207}Pb^*/^{206}Pb^*$ ages are those that have undergone the least amount of ancient Pb loss. Probability density plots and mixture-modelling of $^{207}Pb^*/^{206}Pb^*$ ages (Figure 7) indicate prominent age components at 2749.6 ± 2.5 Ma for Z30 (n = 8/23) and 2745.1 ± 2.2 Ma for Z31 (n = 19/23). Agreement between the oldest groups for the two samples implies that the eight Z30 zircons yielding the oldest group of $^{207}Pb^*/^{206}Pb^*$ ages have indeed experienced mainly recent

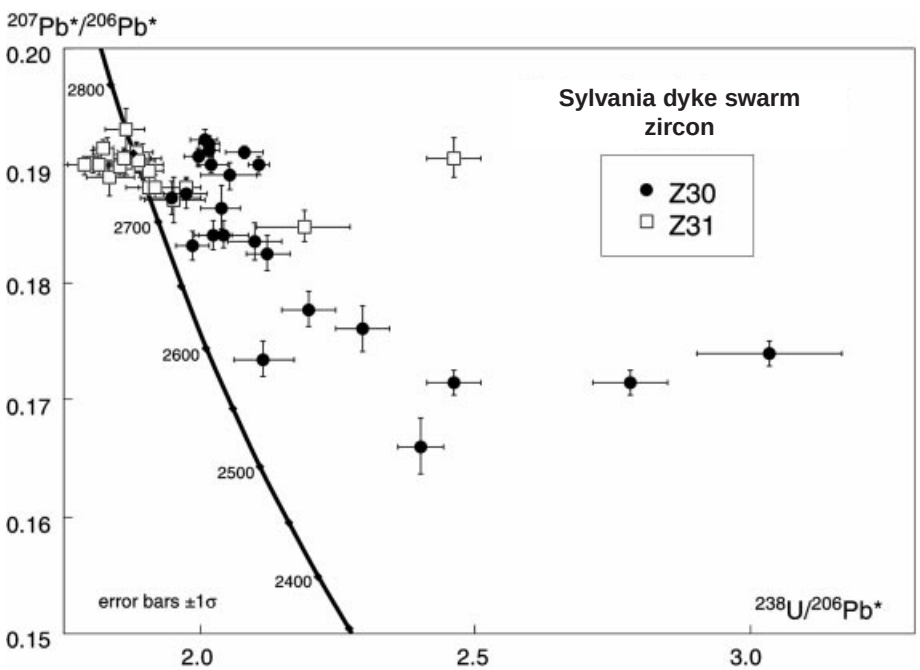


Figure 6 U–Pb evolution diagram illustrating analytical results for zircons from Sylvania dyke samples Z30 and Z31.

Pb loss, and suggests that the two samples are the same age. The possibility that zircons in the Sylvania dykes formed by metamorphic breakdown of primary baddeleyite is made unlikely by the high $^{232}Th/^{238}U$ ratios of the zircons; secondary zircon formed by replacement of baddeleyite typically has very low $^{232}Th/^{238}U$ ratios (≤0.2), similar to those of the original baddeleyite (Wingate *et al.* 1998). The best estimate of the age of the Sylvania dyke swarm is provided by the mean of both results of 2747 ± 4 Ma (95% confidence limits).

DISCUSSION

Several authors have suggested that the Black Range dykes acted as feeders to the Mt Roe flood basalts (Lewis *et al.* 1975; Hickman 1978, 1983; Blake 1993), although this relationship has not been demonstrated in the field. The mean $^{207}Pb^*/^{206}Pb^*$ age of the dykes, 2771.8 ± 1.1 Ma (1σ), is indistinguishable statistically from the mean zircon age of 2770.6 ± 3.5 Ma (1σ) for two samples of Mt Roe Basalt (Arndt *et al.* 1991). Units stratigraphically above the Mt Roe Basalt are unconformable on Black Range dykes (Hickman 1978; Williams 1989) and this prompted Tyler (1990) to speculate that the dykes pre-date development of the Hamersley Basin. The correspondence between their ages (Figure 8), however, implies that the Black Range dykes and the Mt Roe Basalt are comagmatic.

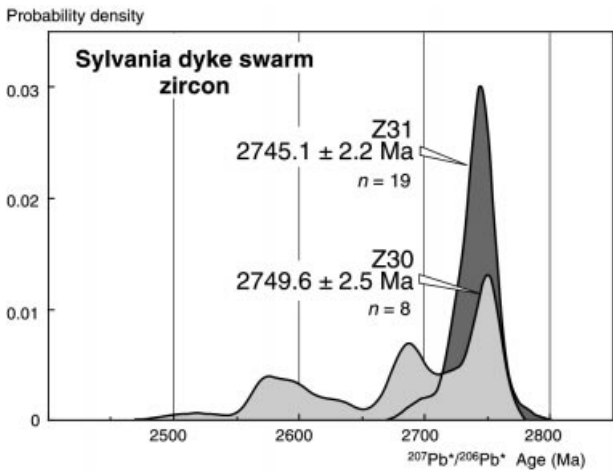


Figure 7 Probability density plot showing the distribution of $^{207}Pb^*/^{206}Pb^*$ ages for zircons from Sylvania dyke samples. Mean ages (±1σ) for the oldest age components for each sample were determined by mixture modelling (Sambridge & Compston 1994).

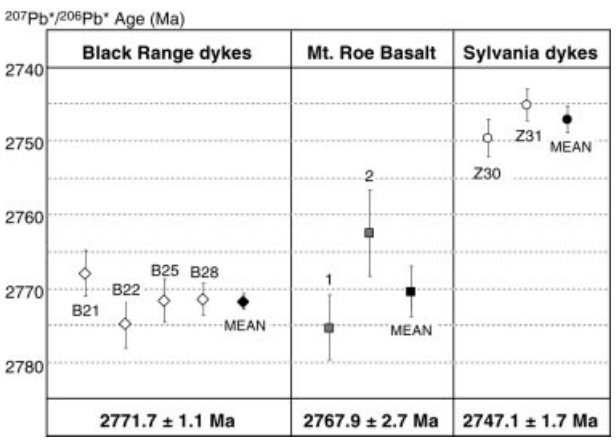


Figure 8 Comparison of $^{207}Pb^*/^{206}Pb^*$ ages for the Black Range dykes, Mt Roe Basalt and Sylvania dykes. Data for the Mt Roe Basalt are from Arndt *et al.* (1991): sample 1, 86738; sample 2, 94792. Error bars and mean ages are ±1σ.

The similar orientations of Black Range dykes, post-Mt Roe Basalt faults, and sedimentary basins and growth faults in the Hardey Formation indicate that craton-wide west-northwest–east-southeast extension continued during deposition of the Hardey Formation (Blake & Barley 1992). It may be worth considering, therefore, as suggested by Blake and Barley (1992) and Blake (1993), that some dykes mapped as belonging to the Black Range swarm may have acted as feeders to subordinate mafic lavas in the Hardey Formation.

Although the $^{207}Pb^*/^{206}Pb^*$ ages of the four Black Range dykes are in good agreement and a precise average can be calculated, their means collectively span an interval of 7 million years. Zircon ages for the Mt Roe Basalt average ~3 million years younger, but their range is greater at 13 million years. This is a relatively long period of time compared to the duration of most younger mafic igneous episodes. Many outpourings of flood basalts occur in less than 5 million years (compilation of Mesozoic and younger events: White & Mackenzie 1989). Conventional U–Pb baddeleyite analyses (LeCheminant & Heaman 1989, 1991) and palaeomagnetic studies (references in Fahrig 1987) show that most, if not all, dykes of the giant 1267 Ma Mackenzie swarm of northern Canada were emplaced within about 2 million years. Two of the dykes dated in this study, the Black Range Dyke (B25) and Cajuput Dyke (B28), yield almost identical ages of 2771.6 and 2771.5 Ma, respectively. These are the two largest of the Black Range intrusions suggesting that the peak of magmatic activity occurred between 2771 and 2772 Ma.

The unconformable contact between the Cajuput Dyke, which is a coarse-grained hypabyssal gabbro, and the overlying sandstone of the Hardey Formation (Hickman 1978; Blake 1993) indicates that considerable uplift and erosion occurred prior to deposition of the sandstone. This is indicated also by a 20 m-thick palaeosol formed by subaerial weathering of Mt Roe flood basalts soon after their eruption (Macfarlane & Holland 1990), and certainly prior to deposition of the overlying Tumbiana Formation (2715 ± 6 Ma: Arndt *et al.* 1991). In addition, fragments and blocks of the Mt Roe Basalt are observed in basal conglomerate of the Hardey Formation (Blake 1984; Blight 1985). Evidence for subaerial weathering and erosion of the Mt Roe Basalt and Black Range dykes, followed by subsidence during deposition of the Hardey Formation, is consistent with the expected response of the lithosphere to the impact of a mantle plume (Campbell & Griffiths 1990; Hill *et al.* 1992; Davies 1994). The Mt Roe Basalt is typical of flood basalts erupted during major rifting of the crust (Myers 1990; Blake & Barley 1992) and hence it is possible that mafic dykes and flood basalts of similar age to the Black Range dykes may be found on other continental fragments that rifted away from the Pilbara Craton at 2.77 Ga.

The age of 2747 ± 4 Ma inferred for emplacement of the Sylvania dykes is 25 million years younger than that determined for the Black Range dykes (Figure 8). It is unlikely, therefore, that dykes in the two areas were emplaced during the same event. The hypothesis that the Black Range dykes 'stitched together' the Sylvania and north Pilbara terranes after their amalgamation (Tyler *et al.* 1992) is not supported by these results (although both terranes are overlain by lower Fortescue Group). Although the Sylvania dykes are not observed to have intruded rocks of the lower Fortescue Group (Tyler 1991), their younger age implies that they may have acted as feeders to volcanic successions that overlie the Hardey Formation (Figure 3). The high zircon content of sample Z31 suggests that a likely candidate for correlation is basalt of the Kylena Formation (or equivalents), samples of which are enriched in Zr (300–470 ppm: Mann 1995).

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SUPPLEMENTARY PAPERS

Table 2 Ion microprobe analytical data for baddeleyite from Black Range dyke samples B21 and B22.

Table 3 Ion microprobe analytical data for baddeleyite from Black Range dyke samples B25 and B28.

Table 4 Ion microprobe analytical data for zircons from samples Z30 and Z31.